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Effects of a Social Skills Intervention on Children with Autism Spectrum Disorder and Peers with Shared Deficits

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Abstract

The current study evaluated the effects of the *Superheroes Social Skills* program (Jenson et al. 2011) in promoting accurate demonstration of target social skills in training and generalization conditions in young children with autism spectrum disorder (ASD) and peers with shared social deficits. Three preschool-age children with ASD and 2 typically developing peers with social deficits attended twice-weekly social skills training over 5 weeks, with lessons targeting nonverbal, participation, requesting, and conversation skills. A multiple probe design across social skills replicated across participants was utilized to determine the effect of participation in the intervention on accurate demonstration of target social skills in training and generalized contexts. Results demonstrated improvements in accuracy of target skill demonstration following introduction of intervention. Parental reports of participant social functioning and parent stress were collected as secondary dependent measures, and also suggest improvements associated with intervention. Limitations and future directions are also discussed.

Keith C. Radley is a coauthor of the *Superheroes Social Skills* program, which was developed for the Utah State Office of Education as part of his doctoral training. Keith C. Radley does not benefit, financially or otherwise, from the commercial sale of the program, nor was the data included in the article part of research funded by the commercial publisher.

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Recent estimates suggest dramatic increases in the prevalence of autism spectrum disorder (ASD), with as many as one in 68 children meeting diagnostic criteria for ASD (Centers for Disease Control and Prevention, 2014). Individuals with ASD experience impairments in social interaction and communication (American Psychiatric Association, 2013), with these impairments being identified as the most salient feature of ASD (Carter, Davis, Klin, & Volkmar, 2005). In comparison to typically developing peers, children with ASD demonstrate fewer social initiations and responses (McConnell, 2002), and exhibit substantial challenges in interpreting social interactions (Jones & Schwartz, 2009). The social deficits exhibited by children with ASD impact their ability to successfully interact with same-age peers and adults (Mason et al., 2014), which results in decreased opportunities to establish friendships and subsequent isolation from peers (Bauminger & Kasari, 2000; Marans, Rubin, & Laurent, 2005; Stewart, Barnard, Pearson, Hasan, & O'Brien, 2006).

Given the prominence of impairments in social interaction and communication in ASD, social skills training is among the most frequently implemented intervention strategies for individuals with ASD (Goin-Kochel, Meyers, & Mackintosh, 2007). Social skills training for individuals with ASD often takes place in a small group format, with groups being composed entirely of individuals with ASD (e.g., Gantman, Kapp, Orenski, & Laugeson, 2012; Sansosti, 2010). Small group instruction may be considered more efficient than one-on-one training, as it provides increased opportunities for observational learning and feedback and allows for similar deficits to be addressed across participants (e.g., Foxx, McMorrow, & Schloss, 1983; Weiss & Harris, 2001). As an example, Palmen, Didden, and Arts (2008) implemented a self-management and performance feedback intervention to address conversation skills in nine adolescents with ASD. Each group was composed of three participants with ASD, with each social skills session consisting of participants role-playing and engaging in table games with the facilitator and other participants. Although social skills interventions that take place in a small group setting and include only individuals with similar disabilities often result in improvements in targeted skills within the training environment, other research suggests that these interventions may be ineffective in producing improvements that are generalized to novel persons or settings (e.g., Bellini, Peters, Benner, & Hopf, 2007; Ganz et al., 2012; Rao, Beidel, & Murray, 2008).

One strategy for enhancing the generalizability of effects of small group social skills for individuals with ASD is the inclusion of

typically developing peers during training (e.g., Chan et al., 2009; National Autism Center, 2009). Interventions incorporating typically developing peers have relied on peer modeling, prompting, and delivery of reinforcement to procedure behavioral change in individuals with ASD, with reviews and meta-analyses indicating the efficacy of such procedures in producing improvements in target behaviors in both training and generalization settings (e.g., Chan et al., 2009; Wang, Cui, & Parrila, 2011; Zhang & Wheeler, 2011). Inclusion of peers in social skills training may promote skill generalization by providing participants with natural consequences of social skill use (i.e., exploitation of current functional contingencies), allowing for incorporation of stimuli from training settings into generalized environments, and training responses to diverse stimuli (Stokes & Osnes, 1989). Although it may be preferable to incorporate only typically developing peers in social skills training for individuals with ASD, it may not always be feasible in clinic or school settings. As such, Sansosti (2010) suggests that peers with similar social deficits who demonstrate basic initiation skills may be considered for inclusion in social skills groups with individuals with ASD. Inclusion of peers with similar deficits provides individuals with ASD with opportunities to practice target skills and receive feedback (Sansosti, 2010), allows for diverse training, provides opportunities for participants to contact natural consequences for social skill use, and may maximize practitioner assets by allowing social deficits to be addressed in a resource-efficient manner, regardless of diagnostic category.

Relatively few studies have evaluated the concurrent effects of social skills training on both individuals with ASD and peers with shared social deficits. Dotson, Leaf, Sheldon, and Sherman (2010) implemented a small group intervention targeting conversational skills of four adolescents with ASD. Additionally, Dotson et al. also included a peer with a diagnosis of ADHD with shared conversational deficits. Participants in the study engaged in an intervention that included didactic instruction, behavioral rehearsal, and performance feedback. Results of the study indicated improvements in conversational skills in three of four participants with ASD, as well as improvements in skill use in the peer with ADHD. Moderate improvements were also noted during generalization probes. In an additional example of social skills addressing shared deficits of individuals with ASD and their peers, Reynolds, Gast, and Luscre (2014) implemented a self-management intervention to promote social initiations in kindergarten students. Participants included two children with ASD, one child with ADHD, and one child with no formal diagnosis who displayed shared deficits. Results indicated that the intervention was effective in increasing

the frequency of initiations of all participants, with improvements also observed in generalized contexts. Although limited, the extant literature indicates that social skills groups including individuals with ASD and peers with shared social deficits may be useful in promoting acquisition and generalization of target social skills.

Superheroes Social Skills

The *Superheroes Social Skills* program (Jenson et al., 2011) is an example of a manualized, multimedia social skills training intervention that incorporates peers in order to promote generalization of target social skills. Whereas many social skills interventions fail to produce generalized effects (e.g., Bellini et al., 2007), *Superheroes Social Skills* explicitly incorporates of strategies identified by Stokes and Osnes (1989) for promoting generalization. Engagement of functional mediators of behavior, or the process of enabling self-control of target behaviors, is incorporated into the program by encouraging self-monitoring of target social skills in training and generalization settings. Training to promote generalization (i.e., presenting multiple stimulus and response exemplars) is done via video models of skill use, behavioral rehearsal with diverse peers, and social narrative comic books. Lastly, exploitation of current functional contingencies is addressed through the inclusion of peers who may provide natural social reinforcement and may serve as discriminative stimuli for skill use in generalized settings.

Although the *Superheroes Social Skills* (Jenson et al., 2011) program is a relatively new intervention, several studies have found the intervention to result in positive outcomes for individuals with ASD. Radley, Ford, Battaglia, and McHugh (2014) investigated the effects of participation in the program on social engagement behaviors during generalized recess periods. Four participants with ASD and three typically developing peers attended the school-based intervention once per week for eight weeks. Although no data were collected for typically developing peer participants, results indicated substantial improvements in the duration of social engagements during recess periods of participants with ASD. In a similar study, Radley, Jenson, Clark, and O'Neill (2014) evaluated the effects of the program on social engagement behaviors when implemented in a clinic setting by trained parents. Five school-age children with ASD and their typically developing siblings participated in weekly social skills training for 8 weeks. Results of the study indicated generalized improvements in the duration of social engagements from training sessions to free play periods in which no intervention was implemented. Pre- and post-measures of parental stress indicated reductions in parent-reported

stress following participation in the intervention, suggesting that targeted social skills may have generalized to novel settings.

The effect of the participation in the *Superheroes Social Skills* (Jenson et al., 2011) intervention on accurate demonstration of target social skills has also been investigated (Radley, O'Handley et al., 2014). Participants in the study, three elementary-age children with ASD, attended a twice-weekly social skills group for 5 weeks. Using a multiple baseline design across skills, four target social skills were taught. Assessment of target social skill accuracy was completed in the training setting. Additionally, generalization of target skills was assessed through novel researchers presenting novel cues for target social skill use. All participants in the study were found to improve in accurate demonstration of target skills following implementation of the intervention, with moderate to strong effects observed in both settings. Additionally, participation in the intervention was associated with improved parent ratings of social functioning for two of three participants, as well as reductions in parent-reported stress for parents of all participants.

Although initial evidence suggests positive outcomes associated with participation in the *Superheroes Social Skills* (Jenson et al., 2011) program, research has focused only on outcomes for school-age participants with ASD and their parents or guardians (Radley, Ford et al., 2014; Radley, Jenson et al., 2014, Radley, O'Handley et al., 2014). As such, it is unknown whether similar improvements in skill use or parental-reported stress would be observed following social skills instruction for preschool-age children with ASD and peers with shared social deficits. It is also important to note that generalization of target skill accuracy in Radley, O'Handley et al. (2014) was assessed via novel researchers, who provided performance feedback following inaccurate skill demonstration. As such, it is unknown whether improvements in skill accuracy would generalize to interaction partners with whom patterns of communication have been previously established (e.g., parents, peers) and who do not provide error correction following instances of inaccurate skill use. The current study was intended to address limitations in assessment of generalization of Radley, O'Handley et al., as well as extend previous evaluations of the *Superheroes Social Skills* program through evaluation of outcomes for young children with ASD and peers with shared deficits. The following research questions were generated to guide the investigation:

1. Does implementation of the *Superheroes Social Skills* (Jenson et al., 2011) program result in improved accuracy of target social skills of preschool-age participants in a training setting during probes delivered by researchers?

2. Does implementation of the *Superheroes Social Skills* (Jenson et al., 2011) program result in improved accuracy of target social skills of preschool-age participants in a generalized setting during probes delivered by parents or guardians?
3. Does implementation of the *Superheroes Social Skills* (Jenson et al., 2011) program result in improved social functioning of preschool-age participants, as rated by parents or guardians of participants?
4. Does implementation of the *Superheroes Social Skills* (Jenson et al., 2011) program result in decreased parent or guardian stress, as reported by parents or guardians of participants?

Method

Participants and Setting

The current study included five participants who were referred to a local university-based clinic for social skills training. Consent for participation was obtained from parents or guardians of each participant after they were given a review of the purpose and procedures of the social skills group. Three participants had been previously diagnosed with ASD by licensed psychologists. At the start of the current study, James was a 5-year, 1-month-old Asian male who had been diagnosed with ASD. He scored in the Very Likely range (Autism Index = 126; 96th percentile) for probability for autism on the Gilliam Autism Rating Scale-Second Edition (Gilliam, 2006) and received intensive behavioral therapy at a separate local clinic. James demonstrated echolalia and stereotyped, repetitive language. He also received supplemental language services at his school under the special education eligibility category of Autism. Ryan was a 5-year-old Caucasian male who had been diagnosed with Pervasive Developmental Disorder-Not Otherwise Specified. On the Autism Spectrum Rating Scales (ASRS; Goldstein & Naglieri, 2010), Ryan's score was in the Slightly Elevated range (T=61) based on parent ratings, and the Very Elevated range (T=70) based on teacher ratings. Ryan received speech and language services at his school through a special education eligibility category of Speech Language Impairment–impaired articulation. No other language deficits were observed. Andrew, a 5-year, 10-month-old Caucasian male, had been diagnosed with mixed receptive-expressive language disorder and autism at the age of 5. He received a total score of 65 on the ASRS, placing him in the Elevated range based on parent ratings. At school, Andrew had a spe-

cial education eligibility category of Autism, and received speech services for articulation. Despite impaired articulation, no other language impairments were observed.

The remaining two participants did not have a current diagnosis of any disorder or disability. Kevin was 4 years, 6 months old at the start of the current study. He was an African American male who had been referred for services by a local Head Start agency for demonstrating conversation skills when interacting with peers. The Behavior and Emotional Screening System (BESS; Kamphaus & Reynolds, 2007) was completed by Kevin's father and Head Start teacher. Kevin received a T-score of 55 when rated by his father, and a 59 when rated by his teacher, indicating normal behavioral and emotional functioning. Although no formal language evaluation was conducted, Kevin demonstrated age-appropriate language skills. Darryl, a 4-year, 6-month-old African American male, had also been referred by the local Head Start agency he attended due to deficits in conversation and requesting skills. On the BASC-2, Darryl received a T-score of 58 when rated by his mother, indicating normal functioning. The BASC-2 completed by Darryl's teacher indicated elevated risk for behavioral and emotional maladjustment (T = 61). Darryl demonstrated age-appropriate language skills.

All social skills training and training setting probes took place in a clinic-based treatment room that was approximately 3 m by 6 m. This room also included a table and four chairs at the center of the room, and a television mounted on a wall. Generalization probes took place in a clinic-based treatment room or a hallway within the clinic building.

Materials

Superheroes Social Skills social narrative comics, self-monitoring cards, and accompanying instructional DVD containing videos of the target skills to be taught (Jenson et al., 2011) were used during the current study. A wall-mounted television was used to display the instructional videos. Play materials (i.e., building blocks, action figures, bouncy balls, and board games) were also used during training and probes.

Measures

Training setting skill probes. The primary dependent variable was participants' accurate demonstration of target skills in the training setting. Three graduate student researchers provided participants with cues to engage in a target social skill using games, toys, and dialog (e.g. "What is your favorite cartoon?" "Ask Mr. Scott if you can play with him," "Get the blue car from Ms. Miner," "Go tell Mr. Vickers what you did this morning"). The skill steps correctly demonstrated

by the participant were recorded using task analyses adapted from the *Superheroes Social Skills* (Jenson et al., 2011) manual (Table 1), and the percentage of steps correctly demonstrated by each participant was then calculated. A minimum of two probes were delivered per session for each target skill assessed in the training setting, with the number of probes administered dependent upon stability of data. The number of probes administered per intervention session ranged from two to six, and varied by participant.

Generalized skill probes. Accurate generalization of target skills across persons was assessed as a dependent variable of the current study. Generalization probes were conducted in a similar manner as training probes, however, parents and guardians administered novel cues for participants to use each target skill. Parents and guardians were used to probe for generalization of target skills as they were thought to represent more naturalistic conversation partners for participants. Similar to training probes, the task analyses presented in Table 1 was used to determine whether each step of the target social skill was completed correctly following delivery of a generalization probe.

Social functioning. In order to identify the target social skills to be targeted during training sessions, each participant's parent or guardian completed the Autism Social Skills Profile (ASSP; Bellini & Hopf, 2007) prior to collection of baseline data. Upon conclusion of the intervention, the ASSP was again completed by parents/guardians to evaluate changes in their report of social functioning. ASSP results include a Total Social Functioning score along with three subscores for Social Reciprocity, Detrimental Social Behaviors, and Participation/Avoidance. Examples items on the ASSP include "interacts with groups of peers," "requests assistance from others," and "maintains eye contact during conversations." A 4-point Likert scale is used on the ASSP, and has raters indicate if an item occurs between a range of never (1) to very often (4). High scores on ASSP scales indicate that the participant engages in more socially appropriate behavior. High internal consistency ($\alpha = .94$) and test–retest reliability ($\alpha = .90$) has been found with the ASSP (Bellini & Hopf, 2007).

Parental stress. The Parenting Stress Index: Short Form (PSI/SF; Abidin, 1995) was completed by a parent or guardian of each participant prior to and following conclusion of social skills training. The PSI/SF was used to evaluate changes in parent stress associated with the introduction of social skills training. Results of the PSI/SF include a Total Stress score and three subscale scores of Parental Distress (PD), Parent-Child Dysfunctional Interaction (P-CDI), and Difficult Child (DC). The PSI/SF has been found to have high internal consistency ($\alpha = .91$), along with a test-retest reliability of .84 (Abidin, 1995).

Table 1
Task Analyses of Target Social Skills

Body Basics	Participation	Wants and Needs	Beginning a Conversation
1. Face the person with head and shoulders oriented toward conversation partner.	1. Demonstrate close physical proximity to interaction partner (within 1 m).	1. Initiate and sustain eye contact with activity partner (3 s).	1. Initiate and sustain eye contact with conversation partner (3 s).
2. Make eye contact within 5 s, and sustain for a minimum of 3 s.	2. Initiate and sustain eye contact with activity partner (3 s).	2. Signal the person you want to talk to (e.g., raise hand, tap on shoulder, say name).	2. State a greeting (e.g., "Hi," "Hello, how are you?").
3. Use appropriate voice with a volume appropriate for the setting.	3. Wait for turn without skipping others or interrupting turn-taking progression.	3. Wait for the person to respond to your signal.	3. Demonstrate listening by sustaining eye contact while other person is speaking (3 s).
4. Use the right expression by having facial expressions match conversation.	4. Join in without disrupting the progression of activity using an activity-appropriate response.	4. Say what is wanted or needed (e.g., "I want . . .").	4. Make a comment or ask a question related to the conversation.
5. Show relaxation, evidenced by relaxed shoulders, even-paced speech, normal breathing rate, and not playing with objects.		5. Wait for person to respond.	
		6. Acknowledge with a reply (e.g., "Okay, thank you").	

Experimental Conditions

Design. A multiple probe across behaviors with concurrent replication across participants design (Cooper, Heron, & Heward, 2007) was used with three experimental phases: baseline, intervention, and maintenance. A new target skill was introduced once 80% of participants demonstrated a minimum of three consecutive probes of 100% for the skill in the training setting. This criterion was implemented as it was believed that performance less than 100% would not suggest meaningful use of the target skill. As the intervention was presented in a group format, a criterion of 80% of participants demonstrating mastery prior to introduction of a new target skill was selected to prevent the non-mastery of one participant from impeding intervention progression when all other participants had demonstrated skill mastery.

Baseline. Upon return of consent forms, parents and guardians of participants completed the ASSP and PSI. The results of the ASSP were then analyzed to identify social skills that were endorsed as deficits across all participants. Items on the ASSP that were scored the lowest, and that prompted selection of target skills from the *Superheroes Social Skills* (Jenson et al., 2011) program were: "Asks questions to request information about a topic," "Joins in a conversation with two or more people without interrupting," "Talks about or acknowledges the interests of others," "Maintains eye contact during conversations," "Maintains the give and take of conversations," and "Interacts with groups of peers." When deficits were identified and matched with skills taught in *Superheroes Social Skills*, baseline skill probes were conducted in the training setting and with parents/guardians for generalization probes. During baseline probes in the training setting, a variety of toys were present to which the participants had free access. Baseline probes in the training setting consisted of delivery of a cue for target social skill use (e.g., "Andrew, come play checkers with me"). Generalization probes during baseline consisted of parent or guardian delivery of similar, yet distinct, cues for social skill use (e.g., "Andrew, come do a puzzle with me). Participants were allowed 5 s to initiate the target social skill, or all steps of the target skill were recorded as incomplete. Praise was provided to participants for compliance to the facilitators' requests to complete tasks during baseline probes, but no performance feedback in relation to accurate skill use was provided. No attention or other social reinforcement was provided except during probes for skill accuracy. Dependent upon data stability, a minimum of three baseline probes were delivered during the initial baseline session, with a minimum of a 30-s interval between probes.

Intervention. Participants took part in 1.5–2 h sessions of social skills training over the course of approximately 5 weeks. Participants attended two social skills training sessions per week. Two to three psychology graduate students facilitated each social skills session. The four target social skills taught to participants, identified using the ASSP, were (a) Body Basics (i.e., nonverbal communication skills); (b) Participation; (c) Wants and Needs; and (d) Beginning a Conversation. Each social skill was presented in a group format, with participants receiving instruction in the same target skill together.

Procedures outlined in the *Superheroes Social Skills* manual (Jenson et al., 2011) guided each training session. At the beginning of each intervention session, the target skill was presented by the facilitators, along with a rationale as to why use of the target skill was important (e.g., it is important to be able to participate so you can play with friends). Next, a video from the *Superheroes Social Skills* DVD was viewed on the television. In each video, animated superhero characters presented the target skill, a rationale for skill use, and four to six discrete steps for target skill demonstration. The last section of the video included video models of unknown, similarly-aged peers successfully demonstrating the skill.

Facilitators then reviewed each step of the skill with participants. The facilitators modeled the incorrect and correct use of the skill and participants were encouraged to identify whether or not the steps had been accurately demonstrated. The facilitators provided error correction if participants were unable to identify correct skill use or missed steps. Participants then engaged in three to five role-plays of the target skill with the facilitators and other participants. During these role-plays, facilitators provided performance feedback if a participant failed to accurately demonstrate all steps of the target skill. Behavior-specific praise was provided to participants for successful skill demonstration. Following role-plays, participants viewed an animated social narrative comic book that showed the superhero characters reviewing the use of the target skill via DVD. Participants were then provided with 10 min of unstructured free play, during which they had access to a variety of toys. At the conclusion of free play, probes of the recently trained skill were administered in the training setting using procedures as described in baseline. However, behavior-specific praise was provided for accurate skill demonstration, and performance feedback was provided for inaccurate demonstration of the skill in intervention. To conclude each session, participants chose a reward (i.e., a small toy or candy) for participation throughout the session. Parents and guardians were also reminded to practice the target skill at home before leaving the training session.

Generalization. Probes of skill generalization for target social skills in baseline were administered before each training session. Probes of skill generalization for target social skills in intervention were administered directly after training sessions. Parents or guardians of participants were provided with cues for target skill use and instructed to deliver the cues to their child. Whereas behavior-specific praise was provided for participation in probes, performance feedback or retraining was not provided following demonstrations of inaccurate skill use during generalization probes.

Maintenance. Previously mastered skills were probed in both the training setting and with generalization partners to assess for skill maintenance periodically over the course of training sessions (approximately 5 weeks). Maintenance probes were conducted directly before training sessions. Baseline procedures for conducting probes were also used during maintenance. Researchers recorded the skill steps correctly demonstrated during each probe, and the percentage of correctly demonstrated steps was calculated. Engagement in probes was followed by praise, delivered by researchers or parents/guardians. No performance feedback was provided following demonstrations of inaccurate skill use.

Interobserver Agreement

Prior to collection of baseline data, observers participated in training sessions with the primary researcher to establish interobserver agreement (IOA). During the training sessions, the primary researcher and observers reviewed task analyses of target skills. Positive and negative examples of target skills were then viewed to practice coding of behaviors. Practice observations were repeated until a minimum IOA of 90% was established between the primary researcher and observers.

A researcher who did not deliver skill probes in the current study calculated IOA. Target social skills were coded independently and simultaneously with the primary observer. IOA was calculated during a minimum of 33% of training and generalization skill probes per phase per participant. IOA was calculated by determining the number of agreements in steps completed, dividing it by the total number of agreements and disagreements, and multiplying the number by 100. Mean overall IOA was calculated to be 99%. Mean IOA for each participant was calculated as follows: Kevin at 99% (range, 96%–100%), James at 96% (range, 85%–100%), Darryl at 99% (range, 98%–100%), Ryan at 99% (range, 90%–100%), and Andrew at 100%. Additionally, Kappa was calculated using the formula $k=(Po-Pc)/(1-Pc)$, where Po is the proportion of agreement between observers and Pc is the proportion

of expected agreement based on change (Uebersax, 1982). Kappa was found to be 0.97 (95% CI = 0.94–1.00) for Kevin, 0.94 (95% CI = 0.88–1.00) for James, 0.96 (95% CI = 0.92–1.00) for Darryl, 0.92 (95% CI = 0.88–0.97) for Ryan, and 1.0 (95% CI = 1.00–1.00) for Andrew.

Treatment Integrity

Treatment integrity was assessed using a treatment implementation checklist derived from the *Superheroes Social Skills* manual (Jenson et al., 2011). Two researchers completed treatment integrity checklists independently and simultaneously for 100% of intervention sessions. Evaluation of treatment integrity checklists found integrity to be 100% for all sessions. IOA for treatment integrity was found to be 100%.

Data Analysis

Visual analysis of level, trend, and variability of the data was conducted to determine the effects of participation in the *Superheroes Social Skills* (Jenson et al., 2011) program on accurate demonstration of target social skills for each participant. Additionally, Nonoverlap of All Pairs (NAP; Parker & Vannest, 2009) was calculated to provide a measure of intervention effect. NAP is a nonparametric method for determining data overlap between each baseline data point and each intervention data point. NAP has been found to be strongly correlated with R^2, and has a statistical power of 91% to 94% of ordinary least squares regression for data conforming to parametric assumptions, and greater than 100% for multimodal data (Vannest, Parker, & Gonen, 2011). Procedures outlined by Parker and Vannest (2009) were used to calculate NAP, but it is also attained as the Area Under the Curve (AUC) from a Receiver Operator Characteristics (ROC) test module. Parker and Vannest suggest that NAP scores should be considered as follows: 0.00 to 0.65 as weak effects, 0.66 to 0.92 as moderate effects, and 0.93 to 1.00 as strong effects.

Results

Accurate Demonstration of Target Social Skills

The primary dependent variable was accurate demonstration of target skills in the training setting, with skill accuracy in a generalized context also assessed as a dependent variable of interest. During baseline, James demonstrated low levels of skill accuracy for all target skills with mean accuracy ranging from 0% to 7% in the training

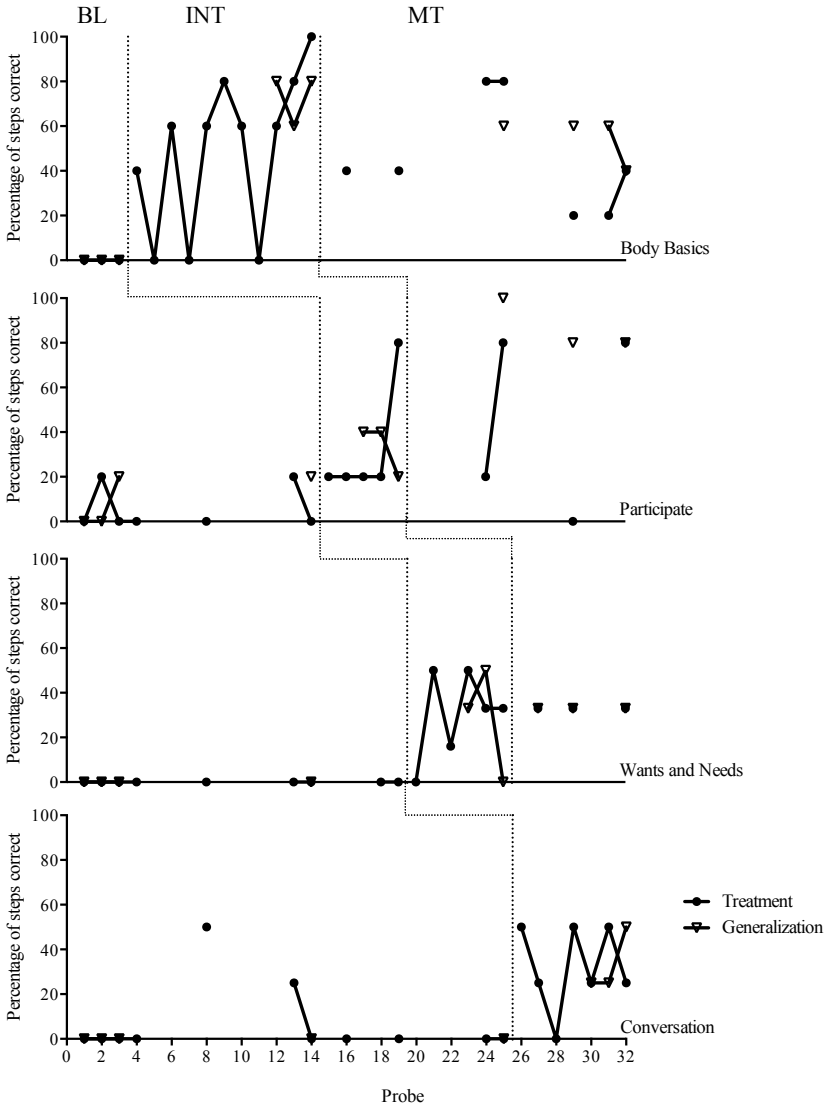


Figure 1. Percentage of steps correctly demonstrated in treatment and generalization settings, James.
Note. BL = baseline; INT = intervention; MT = maintenance.

setting and from 0% to 10% for generalization probes (Figure 1). Following introduction of social skills training, James's demonstrated improvements in skill accuracy for all target skills; however, his results were variable across skills and contexts. In the intervention

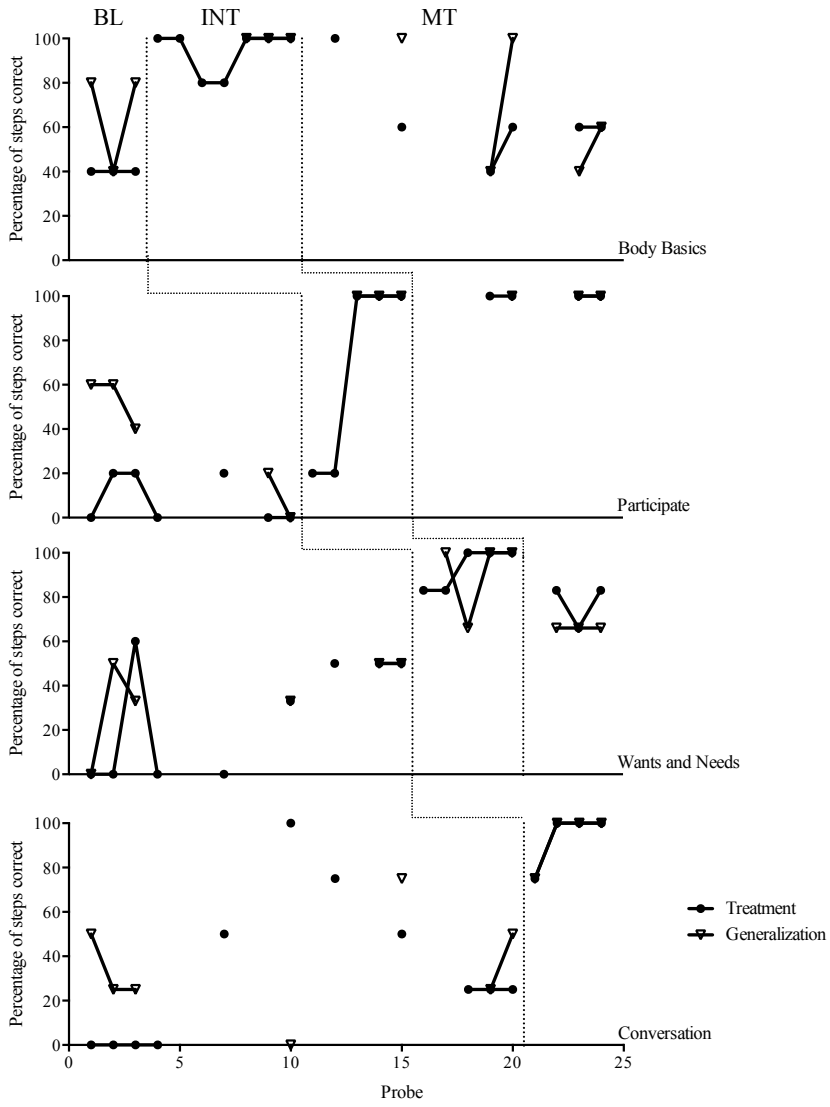


Figure 2. Percentage of steps correctly demonstrated in treatment and generalization settings, Ryan.
Note. BL = baseline; INT = intervention; MT = maintenance.

phase, he demonstrated mean skill accuracy ranging from 30% to 49% in the training setting and from 28% to 73% for generalization probes. Calculation of NAP revealed moderate improvements for all skills in the training setting, and moderate to strong improvements in the

generalized context (Table 2). During the maintenance phase, James demonstrated levels of skill accuracy similar to those demonstrated in the intervention phases. James's mean skill accuracy in maintenance ranged from 33% to 46% in the training setting and from 33% to 87% during generalization probes. NAP calculations for maintenance phase data suggest moderate to strong effects across skills and settings (Table 3).

Ryan's mean skill accuracy during baseline ranged from 8% to 40% in the training setting and from 36% to 67% during generalization probes (Figure 2). Ryan demonstrated improved skill accuracy following social skills training, with decreased variability for most skills. He demonstrated mean skill accuracy during intervention that ranged from 68% to 94% in the training setting and from 92% to 100% for generalization probes. NAP calculations demonstrated strong intervention effects for all skills in training and generalization with the exception of a moderate intervention effect for Participate in the training setting (Table 2). He demonstrated higher levels of accuracy in the training setting compared to baseline levels for all target skills during maintenance probes. Generalization probes revealed maintained improvements for Participate and Wants and Needs skills, whereas Body Basics demonstrated a return to baseline levels. Ryan's mean levels of accuracy during maintenance ranged from 63% to 100% in the training setting and from 66% to 100% in the generalization setting. Additionally, NAP scores revealed strong maintenance effects for all skills with the exception of a moderate effect for Body Basics in the training setting and a small effect for Body Basics in the generalization setting (Table 3).

During baseline, Andrew demonstrated mean skill accuracy ranging from 6% to 23% during training probes and from 0% to 47% during generalization probes (Figure 3). With the introduction of social skills training, immediate increases in level were observed for all target skills with decreases in variability for Body Basics, Wants and Needs, and Conversation skills. Andrew's mean skills accuracy ranged from 80% to 96% for training probes and from 83% to 100% for generalization probes. Calculation of NAP revealed strong intervention effects for all skills in both training and generalization settings, with the exception of a moderate effect for Conversation generalization (Table 2). Maintenance probes conducted in the training setting suggested sustained improvements in accuracy of the Body Basics, Participate, and Wants and Needs skills. Generalization probes during the maintenance phase revealed skill accuracy levels greater than baseline for Body Basics and Wants and Needs skills. Maintenance probes of the Participate skill demonstrated levels similar to baseline in the

generalization setting. Andrew's mean skill accuracy during maintenance ranged from 73% to 88% for training setting probes and from 45% to 85% for generalization probes. NAP calculations demonstrated strong maintenance effects for all skills, with the exception of moderate effects for Participate in both training and generalized contexts (Table 3).

Kevin demonstrated low levels of skill accuracy during baseline for all target skills with variability for the Conversation skill in the generalization setting (Figure 4). Mean skill accuracy ranged from 0% to 20% in the training setting, and from 7% to 50% in the generalization setting. During the intervention phase, Kevin demonstrated immediate increases in levels for Body Basics, Participate, and Conversation skills and increasing trends for the Wants and Needs and Conversation skills. Increased variability was observed in accuracy for Body Basics and Participate skills, with accuracy stabilizing at the conclusion of the intervention phase. Kevin's mean accuracy during the intervention phase ranged from 70% to 86% in the training setting and from 75% to 100% during generalization probes. Calculation of NAP revealed strong intervention effects for Body Basics and Conversation in the training setting, and for Participate and Wants and Needs during generalization probes (Table 2). All other intervention phase NAP calculations were interpreted as moderate effects. During the maintenance phase, probes revealed sustained intervention effects for all skills, with mean accuracy ranging from 73% to 94% in the training setting and from 83% to 87% in the generalization setting. NAP calculations for maintenance phase data revealed strong intervention effects for all skills, with the exception of a moderate effect for generalization probes of Body Basics (Table 3).

For Darryl, data were only gathered on the Body Basics, Wants and Needs, and Conversation skills as a result of absence from the group during training for the Participate skill. Skill accuracy was low for the three target skills presented with means ranging from 0% to 5% in the training setting and 0% to 15% in the generalization setting (Figure 5). Following the introduction of social skills training, Darryl demonstrated immediate increases in level for all target skills. Mean skill accuracy in the intervention phase ranged from 80% to 91% in the training setting and from 75% to 100% in the generalization setting. Calculation of NAP for intervention phase data revealed strong effect sizes across skills and contexts (Table 2). During the maintenance phase, probes revealed sustained intervention effects for Body Basics and Wants and Needs skills in the training and generalization settings. Mean skill accuracy in the maintenance phase ranged from

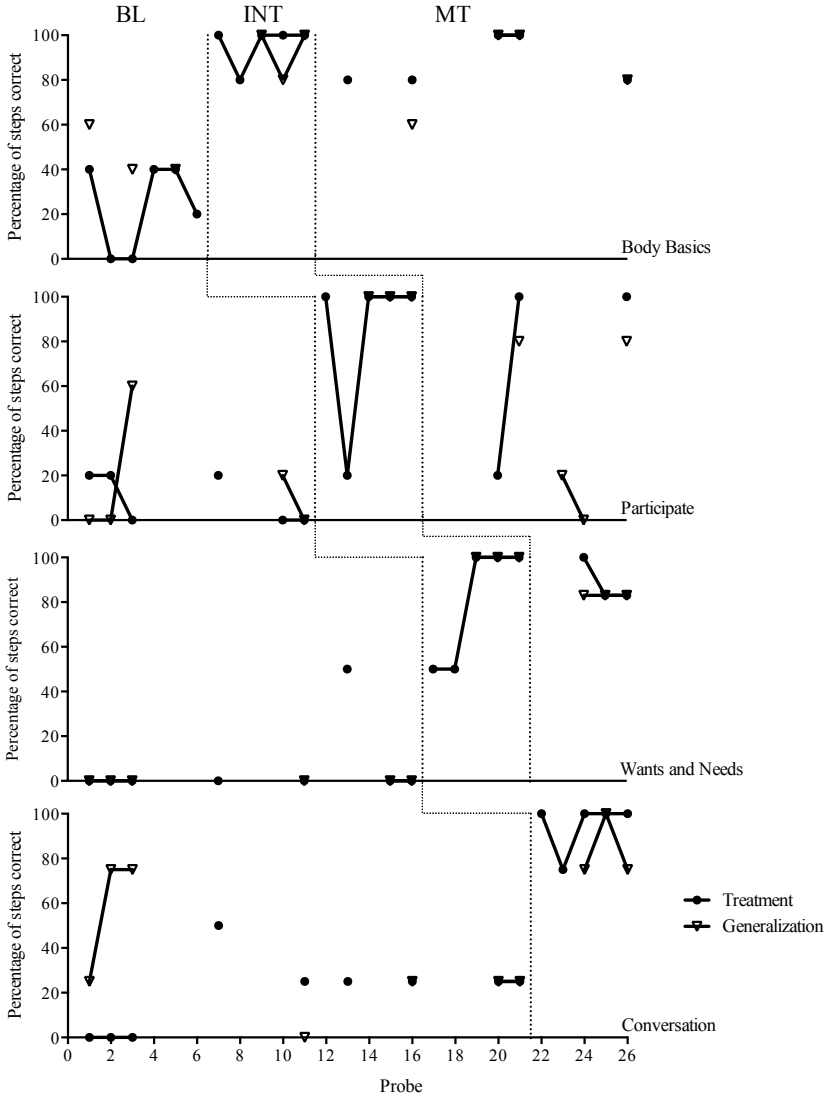


Figure 3. Percentage of steps correctly demonstrated in treatment and generalization settings, Andrew.
Note. BL = baseline; INT = intervention; MT = maintenance.

72% to 78% in the training setting and from 85% to 89% for generalization probes. Additionally, NAP calculations demonstrated strong effects for all target skills in the maintenance phase across training and generalization probes (Table 3).

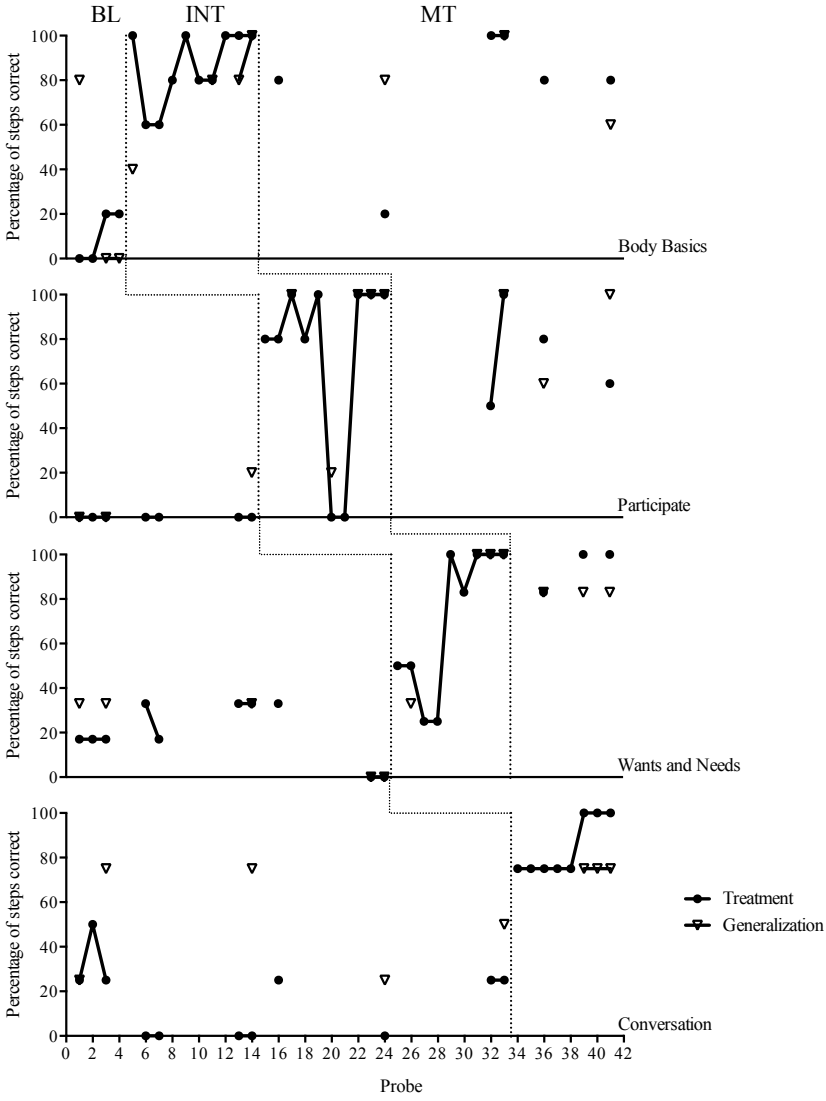


Figure 4. Percentage of steps correctly demonstrated in treatment and generalization settings, Kevin.
Note. BL = baseline; INT = intervention; MT = maintenance.

Social Functioning

Participants' social functioning was measured by having parents or guardians complete the ASSP prior to and immediately following conclusion of intervention. Pre- and post-intervention Total Social

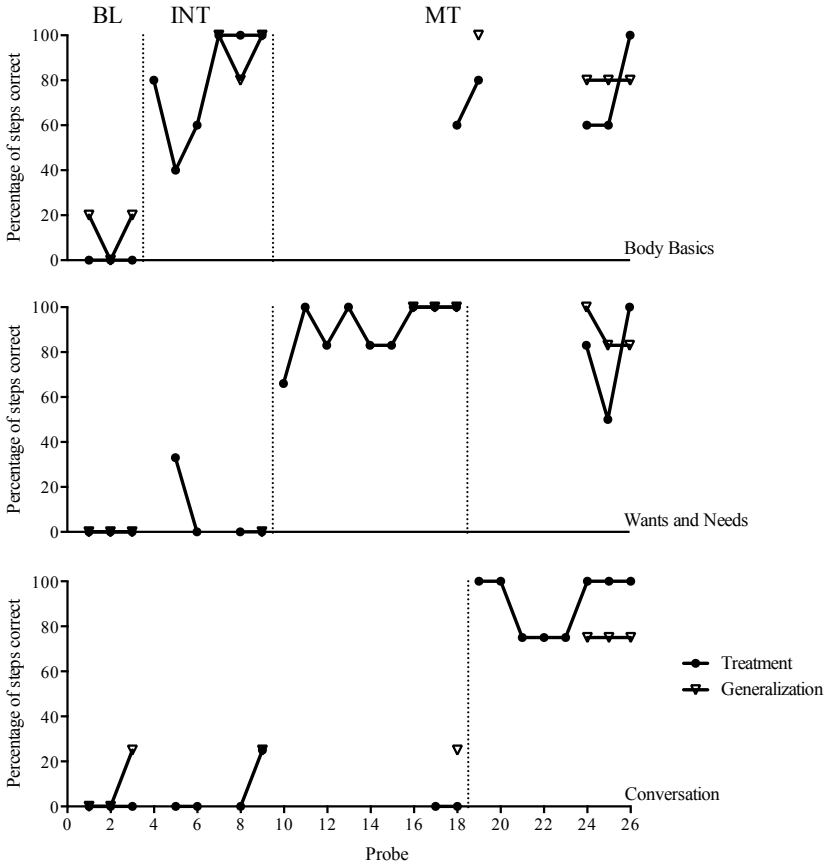


Figure 5. Percentage of steps correctly demonstrated in treatment and generalization settings, Darryl.
Note. BL = baseline; INT = intervention; MT = maintenance.

Functioning scores and subscale scores are presented in Table 4. Improvements in Total Social Functioning were observed for all participants. With the exception of Kevin and Darryl, ratings of Social Reciprocity and Participation/Avoidance improved at post-intervention. Kevin's Detrimental Social Behaviors were rated as improving, whereas other participants demonstrated minimal change.

Parental Stress

The PSI/SF was used to assess stress levels preceding and following the introduction of social skills training. Pre- and post-

Table 2

Skill Accuracy in Training and Generalization, Intervention Phase

	James	Ryan	Andrew	Kevin	Darryl
Body Basics Training	0.86 (0.09–1.37)	**1.00** (0.31–1.69)	**1.00** (0.39–1.60)	**1.00** (0.42–1.58)	**1.00** (0.29–1.71)
Body Basics Generalization	**1.00** (0.16–1.84)	**1.00** (0.16–1.80)	**1.00** (0.16–1.84)	0.83 (–0.11–1.44)	**1.00** (0.16–1.84)
Participate Training	0.89 (0.19–1.35)	0.91 (0.25–1.41)	**0.95** (0.29–1.50)	0.90 (0.32–1.28)	N/A
Participate Generalization	0.92 (0.06–1.61)	**1.00** (0.26–1.74)	**1.00** (0.26–1.74)	**0.97** (0.19–1.67)	N/A
Wants and Needs Training	0.92 (0.32–1.35)	**1.00** (0.45–1.55)	**0.98** (0.39–1.51)	0.91 (0.38–1.27)	**1.00** (0.50–1.49)
Wants and Needs Generalization	0.83 (–0.11–1.44)	**1.00** (0.36–1.64)	**1.00** (0.29–1.71)	**0.93** (0.18–1.52)	**1.00** (0.23–1.78)
Conversation Training	0.84 (0.24–1.15)	**0.93** (0.29–1.45)	**1.00** (0.45–1.55)	**1.00** (0.55–1.45)	**1.00** (0.53–1.48)
Conversation Generalization	**1.00** (0.26–1.74)	**0.98** (0.34–1.59)	0.90 (0.12–1.49)	0.80 (–0.14–1.34)	**1.00** (0.26–1.74)

Note. NAP effect size scores below 0.65 are considered small, scores between 0.66–0.92 are considered moderate, and scores ranging from 0.93–1.00 are considered strong (Parker & Vannest, 2009). Strong effects in **bold**. Ninety % confidence intervals in parentheses.

Table 3

Skill Accuracy in Training and Generalization, Maintenance Phase

	James	Ryan	Andrew	Kevin	Darryl
Body Basics Training	**1.00** (0.31–1.69)	0.92 (0.13–1.54)	**1.00** (0.39–1.60)	**0.96** (0.27–1.48)	**1.00** (0.26–1.74)
Body Basics Generalization	**1.00** (0.23–1.78)	0.53 (–0.67–0.80)	**0.96** (0.14–1.69)	0.83 (–0.17–1.50)	**1.00** (0.23–1.78)
Participate Training	0.80 (-0.02–1.23)	**1.00** (0.38–1.62)	0.92 (0.23–1.54)	**1.00** (0.38–1.62)	N/A
Participate Generalization	**1.00** (0.23–1.78)	**1.00** (0.26–1.74)	0.75 (–0.17–1.17)	**1.00** (0.16–1.84)	N/A
Wants and Needs Training	**1.00** (0.34–1.66)	**1.00** (0.34–1.66)	**1.00** (0.33–1.67)	**1.00** (0.35–1.65)	**1.00** (0.31–1.68)
Wants and Needs Generalization	**1.00** (0.23–1.78)	**1.00** (0.29–1.71)	**1.00** (0.29–1.71)	**1.00** (0.26–1.74)	**1.00** (0.23–1.78)

Note. NAP effect size scores below 0.65 are considered small, scores between 0.66 and 0.92 are considered moderate, and scores ranging from 0.93 to1.00 are considered strong (Parker & Vannest, 2009). Strong effects in **bold**. Ninety % confidence intervals in parentheses.

Table 4
Pre- and Post-Intervention ASSP Scores

	James		Ryan		Andrew		Kevin		Darryl	
	Pre	Post	Pre	Post	Pre	Post	Pre	Post	Pre	Post
Social Reciprocity	39	45	53	62	49	64	58	57	54	53
Detrimental Social Behaviors	20	21	23	27	35	37	20	27	33	33
Participation/ Avoidance	39	45	28	32	26	35	35	32	35	37
Total Social Functioning	98	103	114	131	122	148	123	128	133	135

Table 5
Pre- and Post-Intervention PSI/SF Raw Scores

	James		Ryan		Andrew		Kevin		Darryl	
	Pre	Post	Pre	Post	Pre	Post	Pre	Post	Pre	Post
Parental Distress	45	44	30	29	38	32	57	36	32	25
Parent-Child Dysfunctional Interaction	36	35	25	32	32	25	27	36	23	22
Difficult Child	42	43	36	31	43	36	48	39	30	25
Total Stress	123	122	104	91	122	106	132	111	85	72

intervention scores for Total Parental Stress and subscales Parental Distress (PD), Parent-Child Dysfunctional Interaction (P-CDI), and Difficult Child (DC) are presented in Table 5. Overall, reductions in Total Stress, Parental Distress, and Difficult Child subscales were noted for each parent or guardian. With the exception of Kevin, minimal change was observed on the Parent-Child Dysfunctional Interaction subscale.

Discussion

The present study aimed to evaluate the effects of participation in the *Superheroes Social Skills* (Jenson et al., 2011) program on accurate demonstration of target social skills by preschool-age children with ASD. Participants with ASD demonstrated increases in level and trend

of target skill accuracy from baseline to intervention phases in the training setting, with NAP scores ranging from moderate to strong effects. These results are consistent with previous research demonstrating improvements in target skill accuracy for school-age participants with ASD following social skills training using the *Superheroes Social Skills* program (Radley, O'Handley et al., 2014).

Maintenance data for participants also indicated moderate to strong effects for skill accuracy in the training setting. Despite moderate to strong maintenance effects, it is important to note substantial variability in maintenance phase data of the preschool-age participants in the current study compared to school-age participants in Radley, O'Handley et al., indicating the need for ongoing instruction and review of target social skills for younger children with ASD.

Two of three participants with ASD included in the study demonstrated rapid mastery of all target social skills in the training setting. James, although demonstrating improvements in target skill accuracy, failed to demonstrate mastery of any of the trained skills. This may have occurred for several reasons. One potential reason for the observed difference in skill accuracy may be James' restricted language development (e.g., use of stereotyped language, echolalia). James demonstrated the most dramatic improvements in Body Basics, which did not require use of language. For the remaining skills, which required the use of novel language, James demonstrated smaller improvements. Although preliminary, this finding suggests a minimum level of language development for best outcomes associated with participation in the *Superheroes Social Skills* (Jenson et al., 2011) program.

The current study also addressed limitations of Radley, O'Handley et al. (2014), who assessed generalization via novel researchers who provided error correction following inaccurate skill demonstration. In generalization probes delivered by interaction partners who did not provide any error correction, participants with ASD demonstrated increases in level and trend of skill accuracy from baseline to intervention. With the exception of Body Basics skill for Ryan, substantial improvements in accurate skill use were observed during maintenance phase generalization probes. Similar to maintenance phase probes in the training setting, maintenance phase generalization data demonstrated increased variability compared to Radley, O'Handley et al., again suggesting the need for ongoing instruction and support in target social skills for young children with ASD. Despite findings of variability in skill accuracy during the maintenance phase, findings of generalization of skill accuracy to communication partners with whom patterns of communication have been previously established are notable, as prior research has indicated that social

skills training does not often result in generalized improvements in social skill use (Bellini et al., 2007; Ganz et al., 2012).

Social skills training often occurs in small groups with only children with ASD or with children with ASD and typically developing peers; however, it has been suggested social skills groups including individuals with ASD and peers with shared deficits may benefit all participants (Sansosti, 2010). Such social skill groups allow all participants to contact natural consequences of social skill use and receive diverse training. Groups comprised of individuals with ASD and peers with shared deficits may also allow participants with ASD increased access to peers who demonstrate basic social initiations, which may not be available should training occur only with participants with ASD Therefore, the present study aimed to extend the current social skills training literature by evaluating the effects of the *Superheroes Social Skills* (Jenson et al., 2011) program on accurate skill demonstration of peers with shared social deficits as participants with ASD. Participation in the intervention resulted in increases in level and trend of accurate skill demonstration for peers with shared social deficits from baseline to intervention and maintenance phases. Effect sizes ranged from moderate to strong across all target skills. Furthermore, similar improvements were observed during generalization probes with the peers' parents presenting novel stimuli for target skill use. Therefore, the current study provides additional evidence to support the use of social skills training to simultaneously address the social behavior of children with ASD and peers with shared social deficits in both training and generalization settings, which is consistent with previous research (e.g., Dotson et al., 2010; Reynolds et al., 2014).

The current study provides important information regarding the utility of manualized interventions such as *Superheroes Social Skills* (Jenson et al., 2011). Although mastery of target social skills was observed for four of five participants, accuracy of target skills during maintenance was variable. Such variability in maintenance data, as well as the failure of James to reach mastery criterion during intervention, may indicate the limits of manualized interventions for some participants. In order to maximize maintenance effects, ongoing instruction and customized individual intervention may be necessary and should be considered for individuals who fail to respond to manualized group intervention.

It is notable that all participants demonstrated improvements in their Total Social Functioning score following social skills training as rated by parents of the participants. Improvements were also seen in Social Reciprocity scores for three of five participants, Participation/

Avoidance scores for four of five participants, and Detrimental Social Behaviors scores for three of five participants. Improvements in social functioning are consistent with previous research indicating parental report of clinically significant improvements in social functioning in non-training settings following participation in social skills training (Radley, Ford, et al., 2014; Radley, O'Handley et al., 2014).

The current study also evaluated the effect of child participation in social skills training on parent-reported stress. Checklists revealed clinically meaningful reductions in overall parental stress for parents of Ryan, Andrew, Kevin and Darryl. Additionally, these participants demonstrated reductions in scores on the Difficult Child subscale. Results indicate that improvements in social behavior of the participants in generalized settings (e.g., home) may be associated with reductions in parental stress, supporting previous findings regarding the effect of improved social functioning on parent-reported stress (e.g., Radley, O'Handley et al., 2014).

Results of the current study should be considered in light of several limitations. First, no long-term follow-up data were collected. As such, it is unknown whether improvements in social skill accuracy, social functioning, and parent stress were maintained over time. Future research should conduct maintenance probes of skill accuracy, as well as collect social functioning and parental stress data, after a longer period of time following social skills training. Additionally, increased variability in maintenance of skill accuracy was observed in comparison to previous investigation of the intervention in school-age populations (Radley, O'Handley et al., 2014). Future research may consider evaluation of supports for skill maintenance for preschool-age children with ASD. Second, the *Superheroes Social Skills* (Jenson et al., 2011) program incorporates multiple strategies to promote change in social skill usage. As the current study did not attempt to conduct a component analysis of the program, it is not possible to determine which components of the program were most closely associated with improvements in accuracy of target social skills. Future research should consider conducting a component analysis of the *Superheroes Social Skills* program to determine which components are necessary for behavioral change.

Third, due to the group nature of the intervention, all participants received training of target skills in the same order. As order effects may have influenced skill acquisition data, future research should evaluate if training skills in a particular order is necessary for skill acquisition. In the current study, parents and guardians were asked to practice target social skills at home with their child. However, there was no evaluation of home-based practice of target skills. As

such, it is unknown what effect this may have had on outcomes observed in the current study. Future researchers may consider addressing this limitation through continuous parent or guardian feedback, or the omission of encouragement to engage in practice of target skills while at home. Additionally, the current study relied on parent report to assess changes in social functioning in home and community settings. As no probes of skill accuracy were conducted in these settings, the degree of skill accuracy demonstrated by participants in these settings is unknown. Future research may address this limitation through collection of probe data in a wider variety of generalized contexts. Lastly, baseline probes in generalization indicated greater accuracy than probes in the training setting for several participants. Although no participant demonstrated mastery of target skills during generalization probes during baseline, this finding may indicate greater accuracy during social interactions with partners with whom patterns of communication and reinforcement histories were established. Future research should evaluate variables that may contribute to suppressed accurate responding in the training setting during baseline phases.

Conclusion

Although Sansosti (2010) suggested that small group social skills training may be considered to simultaneously address social deficits of children with ASD and peers with shared social deficits, relatively few studies have evaluated the concurrent effects of social skills training on children with and without ASD. The results of the current study provide support to the limited research finding small group social skills interventions for children with ASD and peers with shared social deficits to effect behavioral change in training and generalized contexts (e.g., Dotson et al., 2010; Reynolds et al., 2013). The current study also extended upon Radley, O'Handley et al. (2014), finding the *Superheroes Social Skills* (Jenson et al., 2011) curriculum to be useful in promoting accurate skill demonstration in preschool-age children and reducing child-related stress of parents and guardians of preschool-age participants. Additionally, the current study addressed limitations of previous research, finding improved accuracy of target social skills in probes delivered by natural interaction partners who provided no performance feedback—further suggesting the utility of the intervention in promoting behavioral change in generalized contexts.

Although the current study was conducted in a clinic setting, the results likely have implications for other settings, such as preschools or play groups, in which social skills deficits of young children are

frequently addressed. Minimal modifications may be required to transport the program to preschools or playgroups, such as reviewing target social skills throughout the school day or viewing videos on a tablet device to allow for use during recess periods. As the intervention utilized in the current study is manualized, it may be utilized by facilitators with minimal training (Radley, Jenson et al., 2014)—increasing feasibility across settings. Although additional research is needed to evaluate behavior change across a variety of settings and interaction partners, the results of the current study suggest that small group social skills training using the *Superheroes Social Skills* (Jenson et al., 2011) program may represent an efficient means of simultaneously addressing the social skills needs of children with ASD and peers with shared social deficits.

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